

Programming In C-HDOC DAY_15

Problem

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Check whether all digits are even.
 - **E.g. 1:** Number: 12345, Output: All digits not even
 - **E.g. 2:** Number: 24680, Output: All digits are even
- Check whether all digits are prime.
 - **E.g. 1:** Number: 2357, Output: All digits are prime
 - **E.g. 2:** Number: 2379, Output: All digits are not prime

ALGORITHM

Step 1: Start.

Step 2: Declare an integer variable c and a character array str.

Step 3: Read the positive integer as a string and store it in str.

Step 4: Display the menu:

- Choice 1: Check whether all digits are even.
- Choice 2: Check whether all digits are prime.

Step 5: Read the user's choice c.

Step 6: If c = 1:

1. Initialize index = 0.
2. Traverse each digit of str.
3. Convert the character digit into an integer using `str[i] - '0'`.
4. Check whether the digit is odd.
5. If the digit is odd, increment index.
6. After checking all digits:
 - If index = 0, display "**All digits are even**".
 - Otherwise, display "**Not all digits are even**".
7. Go to Step 9.

Step 7: If c = 2:

1. Initialize flag = 0.
2. Traverse each digit of str.
3. Convert the character digit into an integer.
4. If the digit is less than 2, it is not prime.
5. Otherwise, check whether the digit is divisible by any number from itself down to 2.
6. Count the prime digits using flag.
7. After checking all digits:
 - If flag equals the number of digits, display **"All digits are prime"**.
 - Otherwise, display **"Not all digits are prime"**.

Step 8: If the choice is neither 1 nor 2, display **"Invalid choice"**.

Step 9: Stop.

C PROGRAM CODE

```
#include <stdio.h>
#include <math.h>
#include <string.h>
int main()
{
    int c;
    char str[100];
    printf("enter a positive integer no\n");
    scanf("%s", str);
    printf("choice1 is to check wether all the digits are even\n");
    printf("choice 2 is to find wether all the digits are prime\n");
    printf("enter CHOICE number\n");
    scanf("%d", &c);
    switch (c)
    {
        case 1:
        {
            int index=0;
            for(int i=0;i<strlen(str);i++)
            {
                if((str[i]-'0')%2!=0)
```

```

    {
        index++;
    }
}
if(index==0)
    printf("all digits are even\n");
else
    printf("not all digits are even\n");
break;
}
case 2:
{
    int flag=0;
    for(int i=0;i<strlen(str);i++)
    {
        int num=str[i]-'0';
        if(num<2)
        {
            int flag=0;
            break;
        }int br=num;
        while(num>1)
        {
            if(br%num==0)
            {
                flag++;
            }
            num--;
        }
    }
    if(flag==strlen(str))
        printf("all digits are prime\n");
    else
        printf("not all digits are prime\n");
    break;
}

```

```

    }
    default:
    {
        printf("invalid choice\n");
        break;
    }
    return 0;
}
}

```

TEST CASES

Test Case	Input	Expected Output	Actual Output	Result (Pass/Fail)
1	enter a positive integer no 24680 choice1 is to check wether all the digits are even choice 2 is to find wether all the digits are prime enter CHOICE number 1	all digits are even	all digits are even	PASS
2	2357 choice1 is to check wether all the digits are even choice 2 is to find wether all the digits are prime enter CHOICE number 2	all digits are prime	all digits are prime	PASS